

Candidate Name

Centre Number

Candidate Number



ZIMBABWE SCHOOL EXAMINATIONS COUNCIL
General Certificate Of Education Ordinary Level

WOOD TECHNOLOGY AND DESIGN

4059/1

Paper 1 Theory, Graphics and Design

3 hours

SPECIMEN PAPER

Additional materials:

Answer booklet,

Coloured pencils,

Drawing paper (A2),

Metric scale rule, scale of 1:1 or 1:5,

Standard drawing equipment, Scientific Calculator.

INSTRUCTIONS TO CANDIDATES

Write your name, centre number and candidate number in the spaces at the top of this page and on **all** separate answer paper used.

Section A Theory (20 marks)

Answer **all** questions.

Write your answers in the spaces provided on the question paper.

Section B Theory (40 marks)

Answer any **three** questions.

Write your answers on the separate answer booklet provided.

Section C Graphics and Design (Compulsory Question) (40 marks)

Answer Question **16** on the **A2** drawing paper provided.

At the end of the examination, fasten the separate answer booklet and drawing paper securely to the question paper.

INFORMATION FOR CANDIDATES

Marks are given in brackets [] at the end of **each** question or **part** question.

You are advised to spend no longer than **20 minutes** on **Section A**, **1 hour** on **Section B** and **1 hour 40 minutes** on **Section C**.

FOR EXAMINER'S USE

SECTION A	
SECTION B	
SECTION C	
TOTAL	

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SECTION A [20 marks]

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*Answer **all** questions in this section in the spaces provided.*

*You are advised to spend **not more than 20 minutes** on this section.*

- 1** **(a)** State **one** responsibility of a public health inspector in construction industry.

_____ [1]

- (b)** Identify **one** Parliamentary Act that governs health and safety in woodworking factories in Zimbabwe.

_____ [1]

- 2** In local home industries, built-up materials are mainly used because of their advantages over solid timber. Complete the diagram in **Figure 2** to show:

- (a)** stout heart in A,
- (b)** lamin board in B.



A



B

Fig. 2

[2]

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- 3 **Figure 3** shows a common woodworking equipment used in a workshop.

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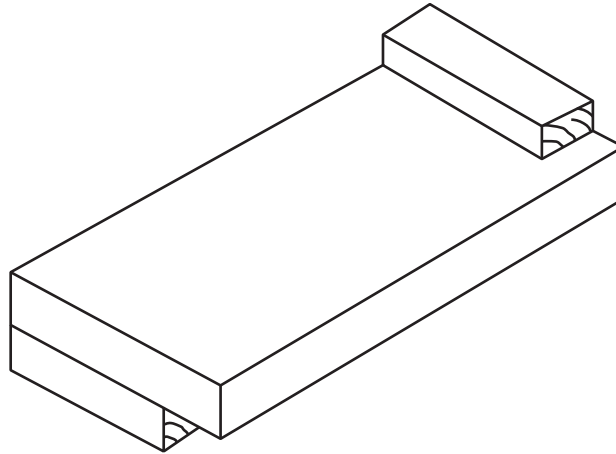


Fig. 3

- (a) Name the equipment shown in **Figure 3**.

 _____ [1]

- (b) Evaluate the use of the equipment shown in **Figure 3**.

 _____ [1]

4 Doors can be classified based on materials used to construct them.

- (a) Identify **one** material apart from timber which can be used to make a door.

 _____ [1]

- (b) On a ledged braced and battened door, state the function of the braces.

 _____ [1]

5 (a) Name **one** type of a coated abrasive paper used to smoothen work before application of a polish.

 _____ [1]

- (b) Explain the use of water when grinding a tool.

 _____ [1]

6 A hand fed surface planner is used for planning the face side and face edge of a work piece.

- (a) State the function of a bridge guard on the surface planner.

 _____ [1]

- (b) Besides planning, name another operation that can be carried out on the surface planner.

 _____ [1]

- 7 (a) Felling timber is done in winter. Evaluate this statement.

 _____ [1]

- (b) Give **one** example of thermoplastics.

 _____ [1]

- 8 **Figure 8** shows drill bits that are generally used when making holes in the workshops.

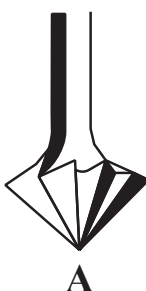


Fig. 8

- (a) Name the bits labelled **A** and **B**.

A _____
 _____ [1]

B _____
 _____ [1]

- 9 **Figure 9** shows an incomplete roof structure of a thatched hut. Label the parts shown **A** and **B**.

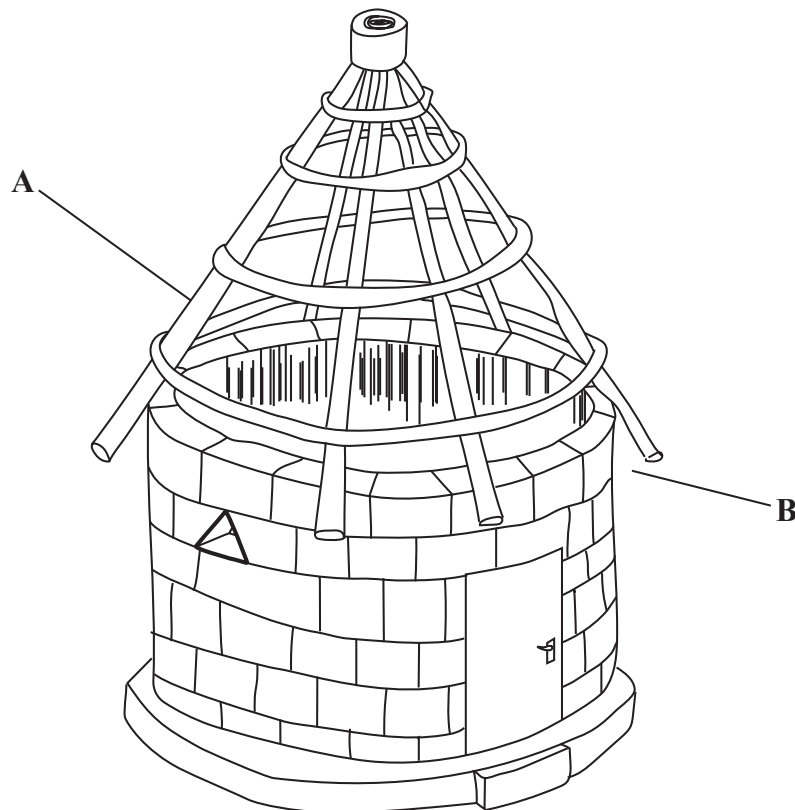


Fig. 9

A _____ [1]

B _____ [1]

- 10 (a) Define recycling in relation to waste management.

 _____ [1]

- (b) Identify **one** disadvantage of incineration as a method of waste disposal.

 _____ [1]

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SECTION B [20 marks]

Answer **three** questions from this section on the separate answer booklet provided.

You are advised to spend **not more than 1 hour** on this section.

- 11** Etula creations is a furniture company in Gokwe that offers a wide range of stylish and functional furniture pieces for homes and offices. The enterprise wants to increase its market share and sales.
- (a) Define the concept of marketing. [2]
 - (b) List **three** marketing strategies that Etula Creations can use to increase its market share. [3]
 - (c) Explain each of the **three** components of the marketing mix that Etula Creations can make use of to boost its sales. [6]
 - (d) Outline any **two** challenges that may be faced when marketing products. [4]
- 12** A number of problems arise when using a recipe book in the kitchen. Pages may turnover, pages may become sticky and dirty while in use.
- (a) Design a wall mounted unit to hold a recipe book while in use. The unit should:
 - (i) allow the recipe book to be read easily.
 - (ii) protect the recipe book from being soiled while in use.
 - (iii) prevent pages from flipping over on their own. [6]
 - (b) Evaluate any **two** suitable factors to be considered when designing the unit. [4]
 - (c) State a suitable finish for the unit. [1]
 - (d) Analyse any **two** suitable material used for making the unit. [4]

- 13 (a) Give **two** purposes of upholstering furniture. [2]
- (b) Identify any **three** methods of managing waste produced during upholstery. [3]
- (c) Using a well labelled diagram, illustrate the **three** main parts of an upholstered chair seat. [6]
- (d) Evaluate the functions of any **two** of the main parts of the upholstered chair. [4]
- 14 **Figure 14** shows a common hardware piece used to secure a door to a door frame.

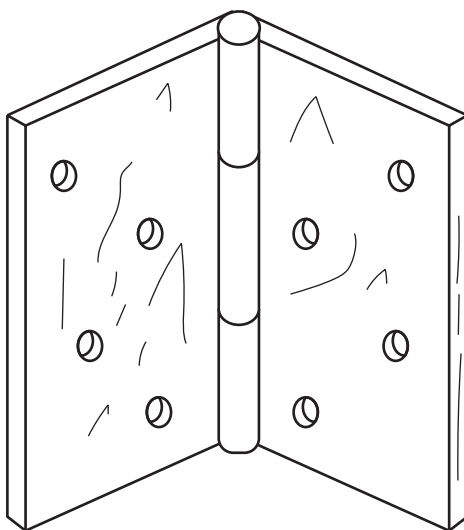


Fig. 14

- (a) Name the type of hardware that is shown in **Figure 14**. [1]
- (b) With the aid of sketches, illustrate how the hardware in **Figure 14** can be fitted onto a door. [4]
- (c) Create a well labelled diagram, to show how two pieces of hardwood timber can be joined using a countersunk head screw. [6]
- (d) State any **two** materials that can be used to make screws. [2]

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15 Mr Moyo, a timber retailer in Mutare sells both hardwood and softwood timber boards.

- (a) State **three** characteristics of hardwood timber board. [3]
- (b) Describe the causes of :
- (i) heart rot, [3]
- (ii) wet rot. [3]
- (c) Illustrate any **three** growth defects in timber. [6]

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Part I (10 marks)

In this part sketches should be free hand.

Answers must be in the space to the right of the vertical line on the prepared drawing paper.

Part I

- (a) Name and sketch an exploded view of the joint used at **A** which allows for a neat appearance from the front. [4]
- (b) Illustrate how the top part of the dresser can be made comfortable for the baby. [2]
- (c) Illustrate how the part labelled **B** could be shaped to improve the appearance and safety. [3]
- (d) Design a suitable wooden handle for the drawer. [2]

Part II

You need to work using a metric scale of **1:5** in this section.

- (a) Draw in either First or Third angle projection.
 - (i) The front elevation in the direction of arrow **F**. [4]
 - (ii) A sectional end elevation of the unit indicated by section **X-X**. Include your design in **Part I**. [10]
- (b) Include the following in your drawing:
 - (i) Four main dimensions to the drawing. [2]
 - (ii) In a suitably sized title block include the following details:
 - scale used
 - projection symbol
 - title
 - quality
 - layout

[4]